WHITE PAPER

LAIRCM INLET/EXHAUST PORT PLUG

Fabrication, Installation, Removal & Maintenance Guide

KC-46 Aircraft | Large Aircraft Infrared Countermeasures System

Originated By	**SSgt Todd R. Hardeman**
Fabrication Lead	SSgt Robert D. Linzy
Organization	Travis Phoenix Spark Manufacturing Team
Testing Location	St. Croix, USVI (SOUTHCOM AOR)
Testing Period	Dec 9, 2025 – Apr 3, 2026 (ongoing)

RECORD OF CHANGES

Rev	Date	Description	Author
1.0	Jan 28, 2026	Initial release – 51-day testing data (Dec 9, 2025 – Jan 28, 2026)	SSgt Hardeman
1.1	Jan 2026	Updated with waterjet design iteration notes	SSgt Hardeman
2.0	Mar 2026	Updated credits, innovation summary, prototype concerns, team composition, testing period extended to Apr 3, 2026	SSgt Hardeman
3.0	Apr 3, 2026	Updated the following: • **2.4** Ease of Replication, 1st sentence. • **4.1** Design Overview • **4.2.1** Early Design • **4.2.2** Current Design • **4.3.1** Equipment and Materials Required • **4.3.2** 3D Printing the Backplate • **4.3.3** Preparing the Silicone Seal	SSgt Linzy
3.1	Apr 5, 2026	Update appendix D team roles.	SSgt Hardeman

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1. EXECUTIVE SUMMARY

This white paper documents the design, fabrication, installation, removal, and maintenance of the LAIRCM (Large Aircraft Infrared Countermeasures) inlet/exhaust port plug for KC-46 aircraft. The plug system was originated by SSgt Todd R. Hardeman, who identified the design flaw, developed the solution, and led field testing - supported by SSgt Robert D. Linzy and the Travis Phoenix Spark Manufacturing Team.

The plug addresses a critical and recurring problem: water intrusion into left and right LAIRCM turret housings causing catastrophic equipment failure at $887,553 per turret replacement. Prior to this initiative, no preventive measure existed - the only response was reactive full turret replacement. The Phoenix Spark team's protective plug represents a fundamental shift from reactive repair to proactive prevention: **a $25 fix for a multi-million-dollar problem**.

A field test was conducted at St. Croix, U.S. Virgin Islands (SOUTHCOM AOR) beginning December 9, 2025, with testing ongoing through April 3, 2026. Data through January 28, 2026 (51 days) yields the following results:

Metric	Result
Turret losses - aircraft WITH plugs (8 turrets)	**0 losses (0%)**
Turret losses - aircraft WITHOUT plugs (4 turrets)	**4 losses (100%)**
Maintainer recommendation rate	**100% (68/68 events)**
Average effectiveness rating	4.18 / 5.00
Cost avoidance (51 days)	**$3,550,211**
Projected annual enterprise savings (fleet-wide)	**$116,713,181**
Projected annual man-hours returned	**370,000 (approx. 835 troops)**
Testing period	Dec 9, 2025 – Apr 3, 2026 (ongoing)

NOTE: Each KC-46 has three LAIRCM turrets: left, right, and aft. The plugs address the left and right turrets only. The aft turret does not experience water intrusion through and does not require a plug.

TESTING STATUS: Testing is ongoing through April 3, 2026. All quantitative data in this document reflects results through January 28, 2026 (51 days). Final results will be incorporated upon test program completion.

2. INNOVATION SUMMARY

2.1 Background and Problem Identified

SSgt Todd R. Hardeman identified a design flaw within the KC-46's Large Aircraft Infrared Countermeasures (LAIRCM) system that posed a significant risk to operational readiness and created enormous financial burden. Unprotected LAIRCM exhaust ports allow water intrusion, causing electrical shorts that necessitate full turret replacement at a cost of $887,553 and 24 hours of maintenance downtime per incident.

In response, SSgt Hardeman & SSgt Linzy engineered and prototyped a low-cost ($25) yet highly durable exterior plug to protect these components while the aircraft is on the ground. Working with the Travis Phoenix Spark Manufacturing Team, the design was refined and submitted through the KC-46 System Program Office (SPO) for field testing approval. With 12 port plugs already deployed to SOUTHCOM, this innovation is on a path toward fleet-wide implementation.

2.2 Level of Innovation

Prior to this initiative, the only procedure was to react to LAIRCM failures by executing a costly and time-consuming replacement of the entire turret assembly. This reactive posture was the accepted norm - no preventive measures existed. The Phoenix Spark team's low-cost protective plug marks a fundamental shift from reactive repair to proactive prevention. By developing a simple, effective solution, the team introduced an entirely new way to protect a critical, high-value asset, fundamentally changing the process from a multi-million-dollar problem to a $25 fix.

2.3 Operational Impact

By preventing LAIRCM system electrical shorts due to moisture, the plug directly increases aircraft availability, ensuring assets are ready for any tasking. The projected return of 370,000 man-hours annually to approximately 835 troops across the fleet allows maintainers to focus on other tasks, significantly boosting unit efficiency and overall readiness. Additionally, the plug directly preserves a vital defense system and ensures the KC-46's lethality is not compromised by an easily preventable environmental factor.

2.4 Ease of Replication

The prototype is made from readily available materials - UV/temperature-resistant Acrylonitrile Styrene Acrylate (ASA), high-temperature silicone, and pre-approved aircraft fasteners - at a cost of approximately $25 per unit. The team secured approval from the KC-46 SPO for field testing. Because the plug is integrated into daily ground procedures as a standard Remove Before Flight item, and each KC-46 base has a fabrication flight, the solution is designed to be replicated across the Air Force using the provided plug blueprints.

2.5 Tangible and Intangible Benefits

- Tangible: Projected fleet-wide annual savings of $116,713,181 in replacement parts costs

- Tangible: Recovery of 370,000 maintenance man-hours annually
- Tangible: $3,550,211 in demonstrated cost avoidance over 51 days of field testing
- Intangible: Empowers Airmen to solve complex operational challenges at the lowest level
- Intangible: Increases mission-capable rate of a tanker fleet essential for force sustainment
- Intangible: Shifts culture from reactive repair to proactive capability preservation

3. BACKGROUND AND PROBLEM STATEMENT

3.1 The Problem

Water intrusion into KC-46 LAIRCM turrets has resulted in catastrophic equipment failure requiring full turret replacement. At Travis AFB, a 10-aircraft fleet averaged 5 turret replacements per year attributable solely to water ingress. Each replacement costs $887,553 in parts, hazardous materials disposal, and labor - and takes 24 hours of maintenance downtime per incident.

In the SOUTHCOM Area of Responsibility, the problem became acute. St. Croix experiences approximately 15 or more rain days per month, creating persistent high-humidity conditions. Within the first three weeks of deployment, two unprotected aircraft (tail numbers 6101 and 6103) suffered complete loss of both their left and right LAIRCM turrets - four turrets destroyed at a combined cost exceeding $3.4 million.

3.2 Root Cause Analysis

Investigation determined that rainwater was entering the left and right LAIRCM turret housings through three exhaust ports per turret. These ports are designed for equipment cooling but become water ingress pathways under wind-driven rain conditions. Contributing factors include:

- Heavy tropical rainfall combined with sustained wind conditions
- Barren flight line environment with minimal windbreaks
- Inlet/exhaust port orientation on left and right turrets creating direct exposure to weather
- Internal moisture control systems overwhelmed under sustained precipitation

3.3 The LAIRCM Turret and Port Layout

Each KC-46 aircraft carries three LAIRCM turrets:

- Left turret - vulnerable to water intrusion; plug required
- Right turret - vulnerable to water intrusion; plug required
- Aft (bottom) turret - different environmental exposure; not affected; no plug required

Each left or right turret has three inlet/exhaust ports. The plug system covers all three ports per turret using a single backplate-and-seal assembly secured to the existing port fasteners.

3.4 Photographic Reference

Figure 1: Close-up view of LAIRCM plug installed on aircraft underside, showing SPARK branding, backplate, silicone sealing surface, fasteners (old, no longer used), and Remove Before Flight streamer.

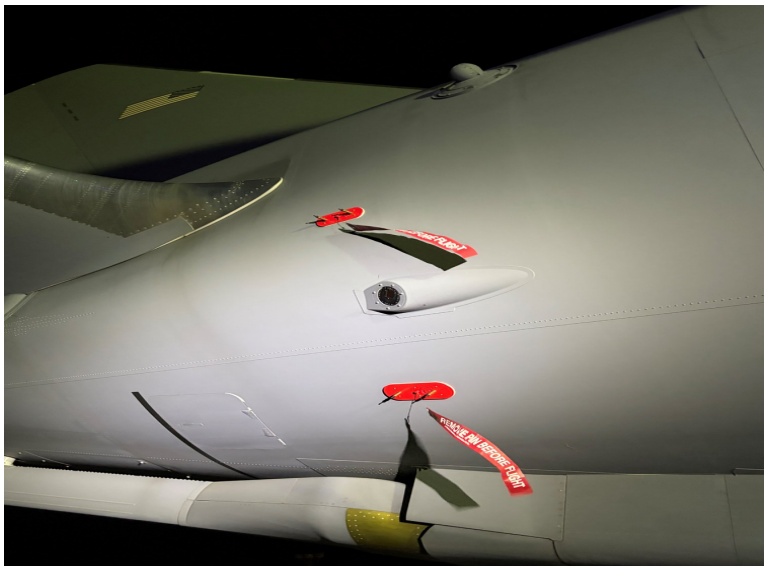

Figure 2: LAIRCM plugs installed on a KC-46 aircraft right side at night, with 'REMOVE BEFORE FLIGHT' streamers clearly visible.

4. PLUG DESIGN AND FABRICATION

4.1 Design Overview

The LAIRCM plug consists of two main components that work together to form a watertight seal over the exhaust ports:

Component	Material	Function
Backplate (Upper)	ASA (Acrylonitrile Styrene Acrylate) – 3D printed	Rigid structural layer that contacts fasteners and provides mounting surface
Backplate (Lower)	ASA – 3D printed	Secondary structural layer, mirror of upper design
Sealing Surface	Hand-poured silicone (molded)	Soft, compliant layer that conforms to port geometry and creates watertight seal
Fasteners	Current: standard hardware (corrosion-prone); Recommended: stainless steel or plunger-type (CLECO)	Secures plug assembly to aircraft exhaust port frame
Safety Streamer	Red fabric with 'REMOVE BEFORE FLIGHT' text	Visual indicator ensuring plugs are removed prior to flight operations

4.2 Design Evolution - Material Research and Trials

4.2.1 Early Design - Dual-Material FDM Printing (ASA + TPU)

The original concept used a combined-material 3D print approach: ASA for the rigid backplate and flexible TPU (Thermoplastic Polyurethane) for the sealing surface. TPU hardness was evaluated from 85A to 95A; even with 2-5% “Line” infill and minimal walls, the material failed to achieve a reliable seal. 60A TPU and silicone at equivalent hardness also failed due to inconsistent print quality. The FDM dual-material approach was discontinued.

4.2.2 Current Design - Hand-Poured Silicone Mold

The current production design separates the rigid backplate (3D-printed ASA) from the sealing surface (hand-poured silicone into a 3D-printed mold). Key parameters:

- Cure time: 7 hours per pour - do not use before full cure
- Sealing performance: reliable and consistent
- Requirement: silicone mold, silicone compound, and 7-hour cure window beyond standard 3D printing
- Additional Consideration: The use of a vacuum chamber is not required but does increase the reliability of the poured silicone by removing excess air from the mix and slightly reduces cure time.

4.2.3 Next-Generation Design - Waterjet Cut Aluminum and Silicone

A new design iteration is in development using a waterjet table to cut aluminum backplates and silicone sealing sheets. This design is intended for higher production rates at any aircraft maintenance fabrication unit with a waterjet table.

4.2.4 Prototype Concern - New Plug Thickness and Conformability

DESIGN CONCERN - ONGOING: The following observations were made by SSgt Hardeman during comparative testing of the new (thicker) vs. old plug design. Development of the perfect prototype is ongoing. The recommendations below are directed to the additive manufacturing team for evaluation.

During comparative testing, the newer, thicker plugs were swapped back to the original plugs on all aircraft except one. The only aircraft that subsequently experienced water intrusion into its port was the aircraft retaining the new plugs. No LAIRCM turret failures were noted, but the intrusion event is significant.

Comparative flexion testing between old and new plugs revealed a critical difference: the new plugs have virtually no bend flexibility, while the old plugs exhibit measurable conformability when hand-flexed. It is theorized that the increased thickness of the new plug body reduces conformance to the aircraft fuselage contour, creating small gaps between the sealing surface and the port frame that allow water to seep in. The thicker plastic body is also much less flexible than the old design.

Additionally, the shorter CLECO fastener expanding jaw assembly on the new plug design is not providing sufficient engagement depth, which may independently compromise seal quality.

Recommendations to the Additive Manufacturing Team:

- Decrease the total body thickness of the new plug to improve conformance to the aircraft fuselage curvature
- Evaluate ASA wall thickness and infill to restore a degree of flexibility comparable to the original plug design
- Procure and evaluate CLECO fasteners with longer expanding jaw assemblies to ensure adequate port engagement depth
- Conduct controlled water spray testing comparing old vs. new plug designs before returning new plugs to service

4.3 Fabrication - Current Production Design

4.3.1 Equipment and Materials Required

Item	Specification / Notes
FDM 3D Printer	Prusa or equivalent; capable of printing ASA material
ASA Filament	UV/temperature-resistant; red color preferred for visibility and Remove Before Flight compliance
Silicone Compound	Pourable two-part high-temperature silicone; 15A Dragon Skin from Smooth-On recommended

Item	Specification / Notes
Silicone Mold	3D-printed ASA mold matching port geometry
Mixing cups and stir sticks	For silicone pour
Mold release agent	Prevents silicone adhesion to mold walls
Fasteners	Stainless steel recommended; CLECO-type with long expanding jaw assembly preferred
Cable	For streamer attachment
Remove Before Flight streamers	Red fabric, standard aviation format
Silicone-compatible adhesive	For bonding seal to backplate; verify cure compatibility

4.3.2 3D Printing the Backplate

1. Obtain the current STEP files for the Upper LAIRCM Port Backplate and Lower LAIRCM Port Backplate.
2. Import STEP files into slicer software (PrusaSlicer or equivalent).
3. Configure print settings:
   - Material: ASA (UV and temperature resistant)
   - Infill: 30% or greater in “gyroid” or another isotropic pattern - structural component
   - Layer height: .2 mm recommended
   - Perimeters/walls: minimum 5 – main strength gaining parameter
   - Ceilings/Floors: Minimum 3 – Infill can be reduced but this parameter will need to be increased to counter the relatively high per square inch force being applied from the fasteners
   - Supports: as required by geometry
   - Color: red ASA preferred for visibility
4. Print Upper and Lower backplate components.
5. Post-process: remove supports, inspect for layer delamination, verify fastener hole dimensions and sealing surface flatness.

4.3.3 Preparing the Silicone Seal

6. Apply mold release agent to interior of silicone mold.
7. Mix silicone compound.
8. Optionally place mix into Vacuum chamber, make sure to only fill mixing container ¾ of the way to allow for expansion.
9. Pour mixed silicone into mold cavity slowly to minimize bubbles.
10. Allow silicone to cure for a minimum of 7 hours at room temperature. Do not disturb during cure.
11. Demold by flexing mold walls to release cured silicone seal.
12. Inspect seal for voids, tears, or incomplete fill. Reject and re-pour if defects present.

WARNING: CURE TIME IS 7 HOURS MINIMUM. Plugs removed from molds before full cure have compromised sealing properties and must not be used.

4.3.4 Assembly

13. Place cured silicone seal against ASA backplate sealing surface.
14. Bond seal to backplate using silicone-compatible adhesive.
15. Install fasteners through backplate fastener holes. Verify correct thread engagement.
16. Attach safety cable through designated anchor point.
17. Attach 'REMOVE BEFORE FLIGHT' streamer to cable. Ensure streamer hangs freely and is highly visible.
18. Perform fit check on known-good port or ground-test fixture before operational use.

5. INSTALLATION PROCEDURES

WARNING: LAIRCM PLUGS ARE FOR GROUND USE ONLY. PLUGS MUST BE REMOVED BEFORE ANY FLIGHT OPERATION.

5.1 When to Install

Plugs shall be installed on the left and right LAIRCM turret inlet/exhaust ports after the aircraft is parked and prior to exposure to rain, high humidity, or overnight conditions.

5.2 Tools and Equipment

- LAIRCM plug set (one larger, two small)
- Plug fasteners

5.3 Pre-Installation Inspection

1. Inspect plug - no cracks, silicone intact, fasteners present, streamer attached and legible.
2. Inspect aircraft exhaust port grates for debris, existing damage, or standing water. Clear as necessary.
3. Verify plug fasteners are present, secure, and free of excessive corrosion.
4. Verify 'REMOVE BEFORE FLIGHT' streamer is attached and in good condition.

5.4 Installation Steps

5. Position plug over the three exhaust ports. Ensure the silicone sealing surface faces the aircraft skin and aligns with port openings.
6. Align backplate fasteners with the corresponding aircraft port fastener receivers.
7. Hand-tighten fasteners in a cross pattern to ensure even sealing pressure. Do not over-torque - this may damage the silicone seal aircraft port.
8. Verify plug is seated flush against the aircraft surface with no visible gaps between the silicone seal and the port frame.
9. Repeat steps 1-5 for the other ports.
10. Document installation in aircraft forms per local MFR requirements.

ALIGNMENT NOTE: The larger plug may require careful alignment to all three ports simultaneously. Verify all three ports are covered before tightening fasteners. Misalignment is a known installation concern - take time to confirm seating before torquing.

5.5 Post-Installation Check

- Visually verify plugs are installed and streamers hang freely
- Confirm fasteners are snug and plug is flush with aircraft surface

6. REMOVAL PROCEDURES

WARNING: PLUGS MUST BE REMOVED BEFORE FLIGHT. VERIFY REMOVAL AS PART OF PREFLIGHT CHECKLIST. Streamers are the primary visual indicator - if streamers are visible, plugs are still installed.

6.1 When to Remove

Plugs must be removed from both the left and right turrets before any flight operation, as part of the pre-flight inspection sequence.

6.2 Removal Steps

19. Loosen fasteners on the plug.
20. Carefully pull plug straight away from the aircraft surface. Do not twist excessively.
21. Inspect plug sealing surface and fasteners immediately after removal.
22. Store plug in designated container (e.g., Pelican case).
23. Repeat steps 1-5 for the opposite turret.
24. Inspect inlet/exhaust ports for any water intrusion, debris, or damage.
25. Document removal in aircraft forms.

6.3 Post-Removal Plug Inspection

- Silicone seal: inspect for tears, compression set, separation from backplate
- Backplate: inspect for cracks, warping, fastener hole wear
- Fasteners: inspect for rust, corrosion
- Streamer: inspect for damage, legibility, secure attachment

Document any deficiencies to Travis Phoenix Spark Innovation Lab. Plugs with significant fastener corrosion, silicone separation, or broken backplates must be removed from service and replaced.

7. MAINTENANCE AND INSPECTION

7.1 Humidity Indicator Monitoring

Indicator Color	Moisture Level	Action Required
BLUE	0% moisture – Optimal	No action required
WHITE	Low moisture present	Monitor closely; inspect for ingress pathway
PINK	High moisture – Concerning	Immediate avionics shop inspection required

With plugs installed on protected turrets throughout the St. Croix test period, all protected left and right turrets consistently maintained more BLUE readings. Unprotected aircraft showed persistent WHITE to PINK readings.

7.2 Fifteen-Day LAIRCM Inspection

Conduct per T.O. DMC-1KC46-A-99-71-1200-07A0A-280A-A_013-02_SX-US:

26. Record humidity indicator readings for all three turrets (left, right, and aft).
27. Inspect LAIRCM filters for evidence of water intrusion.
28. Verify turret assembly condition.
29. Inspect plug condition if currently installed.
30. Document results. Report any WHITE or PINK readings on previously BLUE turrets immediately.

7.3 Fastener Maintenance

Fastener corrosion is the primary maintenance concern identified during testing. In the St. Croix environment, 42% of removal events and 30% of installation events noted fastener rust or corrosion.

RECOMMENDATION: Replace all standard hardware fasteners with stainless steel fasteners or CLECO-type plunger quick-release fasteners with long expanding jaw assemblies. Apply anti-corrosion coating to all metal components. Use rubber or polymer threading to prevent aircraft paint wear.

Issue	Frequency (St. Croix)	Corrective Action
Fastener rust/corrosion	42% removal, 30% installation events	Replace with stainless steel or long-jaw CLECO fasteners
Paint wear at installation points	5 occurrences	Add rubber/polymer coating to fastener threads; do not over-torque

Issue	Frequency (St. Croix)	Corrective Action
Plug misalignment (large plug)	3 occurrences	Improve alignment guides; verify pre-tightening
Wet plugs from storage condensation	2 occurrences	Store in Pelican case with desiccant
Silicone separation from backplate	1 occurrence	Inspect bonding; replace plug if silicone is separating

7.4 Plug Replacement Criteria

Replace a plug immediately if any of the following are found:

- Silicone seal: visible tears, splits, significant compression set, or separation from backplate
- Backplate: cracks, fractures, significant warping, or damaged fastener holes
- Fasteners: stripped threads, severe corrosion, or missing
- Streamer: missing or illegible 'REMOVE BEFORE FLIGHT' text

8. TESTING PROGRAM RESULTS

TESTING STATUS: Testing is ongoing through April 3, 2026. All data below reflects results through January 28, 2026 (51 days). Final metrics will be updated upon test program completion.

8.1 Test Configuration

Group	Aircraft	L/R Turrets	Turrets Lost	Cost Impact
WITH Plugs	6097, 6040, 6094, 6085	8	**0 (0%)**	**$0**
WITHOUT Plugs	6101, 6103	4	**4 (100%)**	**$3,550,211**

8.2 Results Summary Charts

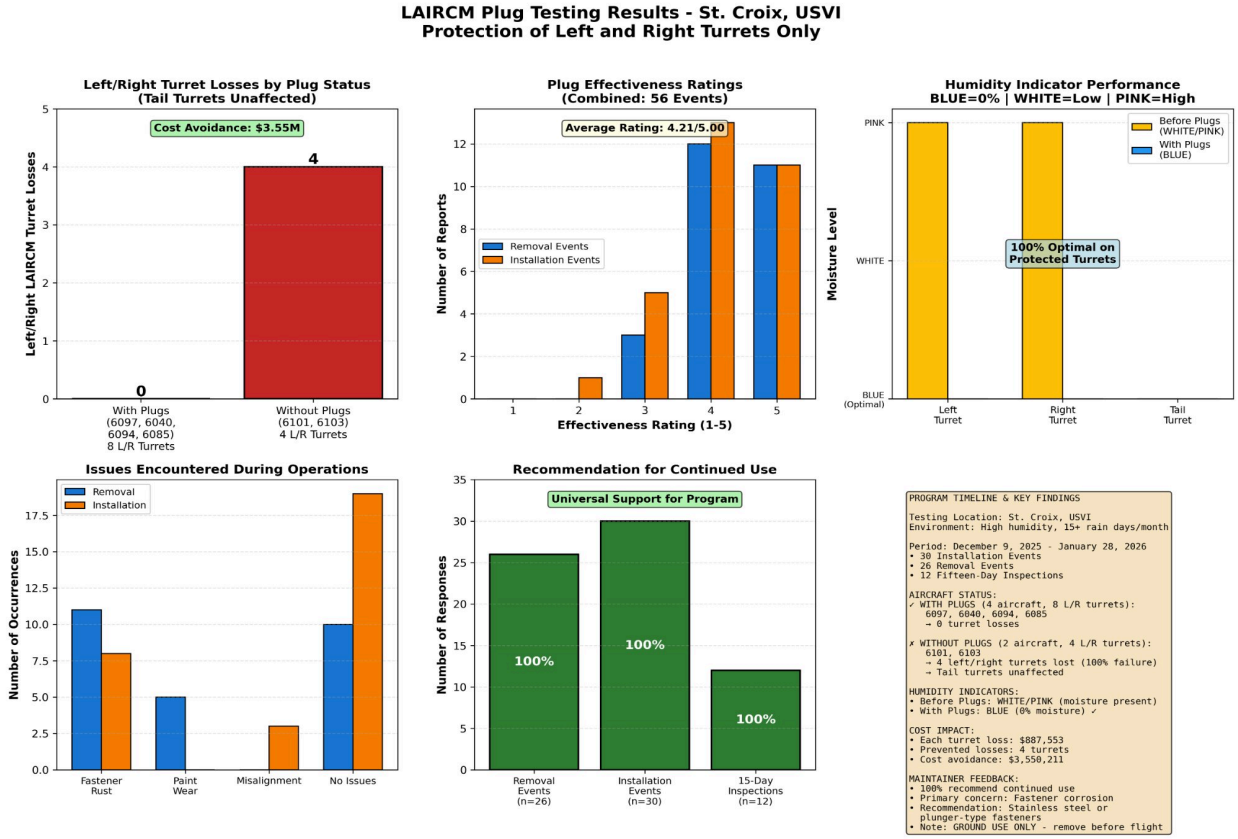

Figure 3: LAIRCM Plug Testing Results – St. Croix, USVI (data through Jan 28, 2026). Charts show turret loss comparison, effectiveness ratings, humidity indicator performance, issues encountered, and recommendation rates.

8.3 Effectiveness Ratings

Event Type	Events	Average Rating (1-5)
Installation Events	30	4.10 / 5.00
Removal Events	26	4.27 / 5.00
OVERALL	**56**	**4.18 / 5.00**

8.4 Maintainer Support

- Installation events: 30 of 30 (100%) recommend continued use
- Removal events: 26 of 26 (100%) recommend continued use
- Fifteen-day inspections: 12 of 12 (100%) recommend continued use
- Total: 68 of 68 events (100%) support program continuation

9. COST ANALYSIS

9.1 Per-Turret Replacement Cost

Cost Component	Amount
LAIRCM Turret Assembly	$850,000.00
Hazardous Materials (HAZ)	$10,552.71
Labor (2x SrA, 1x SSgt)	$27,000.00
TOTAL PER TURRET	**$887,552.71**

9.2 Demonstrated Cost Avoidance – 51 Days

Metric	Value
Turrets lost without plugs	4
Turrets protected with plugs	8
Cost per unit	$887,552.71
Total Cost Avoidance (51 days)	**$3,550,211**

9.3 Enterprise Projection – Annual Savings

Metric	Value
Total USAF KC-46 procurement	263 aircraft
Vulnerable left/right turrets	526 (2 per aircraft)
Mobile bases	11 locations
Projected annual turret losses (at Travis baseline rate)	131.5
Man-hours returned annually	370,000 (approx. 835 troops)
Projected Annual Enterprise Savings	**$116,713,181**

10. RECOMMENDATIONS

10.1 Immediate Actions

Approve Fleet-Wide Implementation

Based on conclusive test results, immediate fleet-wide implementation is strongly recommended with priority given to high-humidity, tropical, and maritime operating locations.

Fastener Upgrade Program

- Replace all current plug fasteners with stainless steel hardware
- Evaluate CLECO-type quick-release fasteners with long expanding jaw assemblies
- Apply anti-corrosion coating to all metal plug components

10.2 Implementation Requirements

Pre-Implementation Inspections

- Avionics shop inspection of all LAIRCM filters for existing water intrusion
- Inlet/exhaust port condition assessment on left and right turrets

Aircraft Forms Documentation

- Add MFR to aircraft forms clearly stating ground-use-only restriction
- Specify installation and removal procedures.

Training and Procedures

- Train all maintainers on proper installation and removal techniques
- Emphasize fastener torque limits - hand-tight only, do not over-torque

11. QUICK REFERENCE CARD

PURPOSE: Job-aid summary for maintainers. For full procedural detail, refer to Sections 5 - 7.

Installation - Quick Steps

31. Inspect plug - no cracks, silicone intact, fasteners present, streamer attached
32. Inspect port - clear of debris and water
33. Align plug over all three exhaust ports of left or right turret
34. Hand-tighten fasteners in cross pattern - do not over-torque
35. Verify flush fit - no visible gaps between silicone and port frame
36. Confirm streamer hangs freely and is visible
37. Repeat for other ports
38. Document in aircraft forms

Removal - Quick Steps

39. Identify both plugs by REMOVE BEFORE FLIGHT streamers
40. Loosen fasteners in cross pattern on first plug
41. Pull plug straight away from aircraft
42. Inspect plug
43. Store in Pelican case
44. Repeat for other ports
45. Inspect ports for water/damage
46. Document in aircraft forms

WARNING: PLUGS ARE GROUND USE ONLY. REMOVE BEFORE FLIGHT.

DO	DO NOT
Remove plugs before all flight operations	Over-tighten fasteners (damages seal and paint)
Inspect plug before each installation	Use a plug with separated silicone or broken backplate
Store plugs in Pelican case	Store plugs loose where condensation can accumulate
Use stainless steel or long-jaw CLECO fasteners	Depart without confirming both plugs are removed

12. CONCLUSION

The LAIRCM inlet/exhaust port plug is a proven, low-cost solution to a significant equipment reliability and cost problem, originating from SSgt Todd R. Hardeman's identification of the design flaw and development of the fix, supported by SSgt Robert D. Linzy and the Travis Phoenix Spark Manufacturing Team. The 51-day operational test at St. Croix, USVI - with testing continuing through April 3, 2026 - has already delivered unambiguous results: four aircraft with plugs protected all eight left and right turrets with zero losses, while two aircraft without plugs lost all four left and right turrets at a cost of $3.55 million.

Maintainer acceptance has been universal. Across 68 test events, 100% of maintainers recommended continued use, and average effectiveness ratings consistently exceeded 4.0 on a 5-point scale. Humidity indicators on protected turrets maintained optimal BLUE readings throughout the testing period, providing objective confirmation of plug performance.

Design refinement is ongoing. The current prototype iteration has identified a conformability concern with the thicker backplate design, and the additive manufacturing team is working to address thickness, flexibility, and fastener engagement depth in the next revision.

With projected enterprise savings of $116.7 million annually across the full KC-46 fleet and $3.55 million in demonstrated cost avoidance over 51 days, the investment in LAIRCM plugs delivers exceptional return. Fleet-wide implementation is strongly recommended with priority given to high-humidity and tropical operating locations.

FINAL RECOMMENDATION

Approve immediate fleet-wide implementation of LAIRCM exhaust port plugs for left and right turrets on all KC-46 aircraft, with priority given to high-humidity and tropical operating environments.

APPENDIX A: TESTING EVENTS SUMMARY

Metric	Count
Total Testing Events (through Jan 28, 2026)	68
Installation Events	30
Removal Events	26
Fifteen-Day Inspections	12
Data Collection Period	51 days (Dec 9, 2025 – Jan 28, 2026)
Full Testing Period	Dec 9, 2025 – Apr 3, 2026 (ongoing)
Turrets Protected (Left/Right)	8 (2 per aircraft x 4 aircraft)
Turrets Lost - With Plugs	0
Turrets Lost - Without Plugs	4 (100% of unprotected)

APPENDIX B: APPLICABLE DOCUMENTS

Document	Description
T.O. DMC-1KC46-A-99-71-1200-07A0A-280A-A_013-02_SX-US	LAIRCM 15-day inspection technical order
Aircraft MFR (Memorandum For Record)	Local documentation for plug installation/removal
LAIRCM_Plug_Testing_Report_FINAL_v2.docx	Full testing program final report, Jan 28, 2026
LAIRCM_Testing_Data_Summary_FINAL.xlsx	Complete raw testing data, all 68 events
Upper_LAIRCM_Port_Backplate.step	3D model file - upper backplate (current production design)
Lower_LAIRCM_Port_Backplate.step	3D model file - lower backplate (current production design)
KC- 46_LARICM_Inlet___Exhaust_Plugs_af1067.pdf	Original plug specification reference document

APPENDIX C: GLOSSARY

Term	Definition
ASA	Acrylonitrile Styrene Acrylate - UV/temperature-resistant thermoplastic used for 3D-printed backplate
CLECO	Quick-release fastener type; plunger/expanding jaw mechanism recommended for plug fastening
FDM	Fused Deposition Modeling - 3D printing method used for backplate fabrication
FOD	Foreign Object Damage - debris or hardware that could damage aircraft or engines
HAZ	Hazardous materials cost component of turret replacement
LAIRCM	Large Aircraft Infrared Countermeasures - self-protection system on KC-46
MFR	Memorandum For Record - aircraft forms documentation entry
SOUTHCOM AOR	U.S. Southern Command Area of Responsibility
SPO	System Program Office - KC-46 program management office that approved field testing
T.O.	Technical Order - official Air Force maintenance instruction document
TPU	Thermoplastic Polyurethane – flexible filament evaluated in early plug design iterations
USVI	U.S. Virgin Islands

APPENDIX D: TEAM COMPOSITION

The following personnel comprise the Phoenix Spark team responsible for the conception, design, fabrication, testing, and documentation of the LAIRCM Exhaust Port Plug program.

Role	Name	Email
Program Originator, SME & Designer	SSgt Todd R. Hardeman	todd.hardeman@us.af.mil
Fabrication Lead & Designer	SSgt Robert D. Linzy	robert.linzy@us.af.mil
Program Lead	Maj Dustin A. Wallace	dustin.wallace.6@us.af.mil
Project Manager	John P. Dickerson	john.dickerson.15@us.af.mil
Project Manager	MSgt Adam C. McDonough	adam.mcdonough@us.af.mil
Project Manager	MSgt Skylar J. Stevens	skylar.stevens@us.af.mil
Team Member	TSgt Charles P. Goley	charles.goley@us.af.mil
Team Member	SSgt Saajaadeen M. Jeffries	saajaadeen.jeffries@us.af.mil
Team Member	SrA Adam W. Nord	adam.nord@us.af.mil

APPENDIX E: NEW vs. OLD PLUG COMPARISON

The following photograph was captured by SSgt Hardeman during comparative testing between the newer (thicker) plug design and the original plug design. The image documents the dimensional differences between the two versions and illustrates the sealing gap concern identified during field observation.

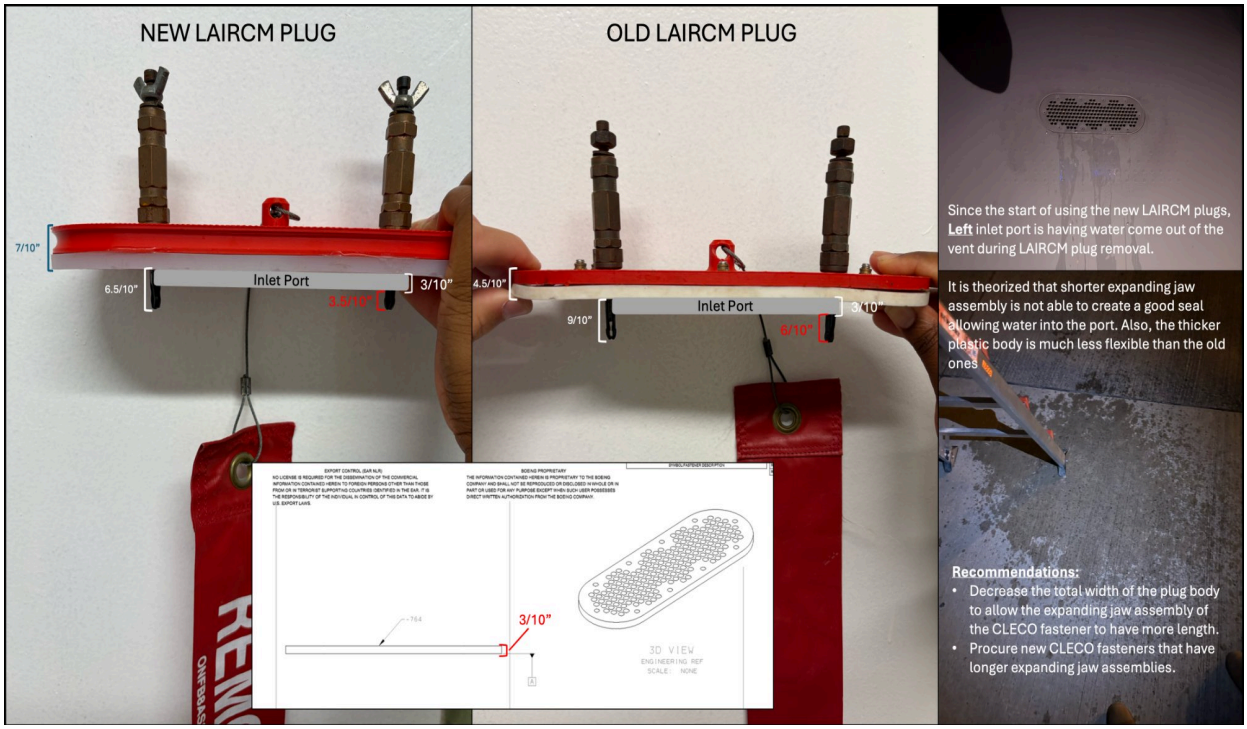

Figure 4: Side-by-side dimensional comparison of NEW (left) and OLD (right) LAIRCM plug designs. Key measurements are annotated. The new plug body is thicker (3.5/10" vs 6/10" silicone depth), resulting in reduced conformability to the aircraft fuselage contour. The shorter CLECO expanding jaw engagement on the new design is also visible.

Key Dimensional Differences Observed

Measurement	NEW Plug	OLD Plug	Concern
Overall body width	6.5/10"	9/10"	New plug is narrower
Inlet port overhang	3/10"	3/10"	Same
Silicone/seal depth below inlet	3.5/10"	6/10"	New plug has shallower seal depth - less conformance
Body thickness (flex)	Rigid - virtually no flex	Measurable hand flex	New plug does not conform to fuselage curvature
CLECO jaw engagement	Shorter - reduced engagement	Longer - adequate engagement	New plug may not achieve positive port seal

DESIGN ACTION REQUIRED: The additive manufacturing team should address plug body thickness reduction, flexibility restoration, and CLECO fastener jaw length before returning the new design to operational service. Refer to Section 4.2.4 for full recommendations.

— END OF DOCUMENT —